Spatio-Temporal Air Quality Forecasting in Urban Sri Lanka: A Comparative Multi-Modal Analysis of Colombo and Kandy

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List of Abbreviations

AOD: Aerosol Optical Depth

BAM: Beta Attenuation Monitor

CEA: Central Environmental Authority (Sri Lanka)

CMC: Colombo Municipal Council

GEE: Google Earth Engine

KMC: Kandy Municipal Council

LSTM: Long Short-Term Memory

MERRA-2: Modern-Era Retrospective analysis for Research and Applications

NO2: Nitrogen Dioxide

PM2.5: Particulate Matter (diameter < 2.5 micrometers)

RMSE: Root Mean Square Error

SHAP: SHapley Additive exPlanations

TROPOMI: Tropospheric Monitoring Instrument

CHAPTER 01: PROBLEM

1.1 Chapter Overview

This chapter is for establishing research content by taking a thorough look at the air pollution crisis in Sri Lanka's two prime cities which are Colombo and Kandy. It is giving light into the shortcomings of the current available monitoring infrastructure which is limited to sparse point-source data in urban centers. This chapter also precisely portraying the **"Spatio-Temporal Gap"** which is the inability for current frameworks to consider the distinct topographic and meteorological challenges of these two primary cities (coastal dispersion vs. valley trapping). And at last this chapter is also about looking into the important related works and formally defines the research problem, outlines the proposed **Hybrid Deep Learning solution** which is focusing on these two target cities.

1.2 Background / Problem Domain

1.2.1 Context: The Tale of Two Cities

The largest environmental health risk in Sri Lanka is air pollution, but the dynamics vary significantly depending on the location and this research focus on the two most important urban environments:

1. **Colombo (Western Province):** The commercial capital of Sri Lanka is Colombo. It has flat coastal terrain, It has high vehicular emissions because it's the most densely populated area on the island, the sea breeze helps to disperse pollution. However, transboundary pollution from India which is carried over by the Northeast Monsoon is impacting the city significantly.

2. **Kandy (Central Province):** Kandy which can be considered as the cultural capital sits in a distinct **"basin" or valley topography**. Despite not having as much traffic, Kandy often is recording higher and more persistent levels of Particulate Matter 2.5 (PM_2.5). This is because of the surrounding mountain ranges which trap the pollutants, preventing it from dispersing.

1.2.2 The "Retrospective" Limitation

Both cities currently rely on **retrospective analysis** for environmental management; the authorities are using historical data to understand past pollution trends. **Ambanwala (2025)** successfully analyzed the pollution trends in these two cities ranging from 1980 to 2024 and it confirmed that the air quality is worsening in Kandy because of the urban density within the valley. But this study offers no way to predict the forecast the air quality. There is a clear lack of **predictive intelligence** that leaves the authorities reactive; they cannot divert traffic or close schools until the damage is done.

1.2.3 The Monitoring Gap

Despite being the most developed cities in Sri Lanka, there isn't adequate monitoring infrastructure,

- **Colombo:** Relies heavily on the US Embassy's reference monitor and a few PurpleAir sensors. The suburbs (e.g., Malabe, Moratuwa) remain largely unmonitored "blind spots."
- **Kandy:** Monitoring is even more fragmented. The complex terrain means a sensor in the city center may not reflect air quality just 2km away on the hillside. This creates a need for a **"Virtual Sensing"** approach—using satellite data to infer air quality across the entire urban footprint of both cities.

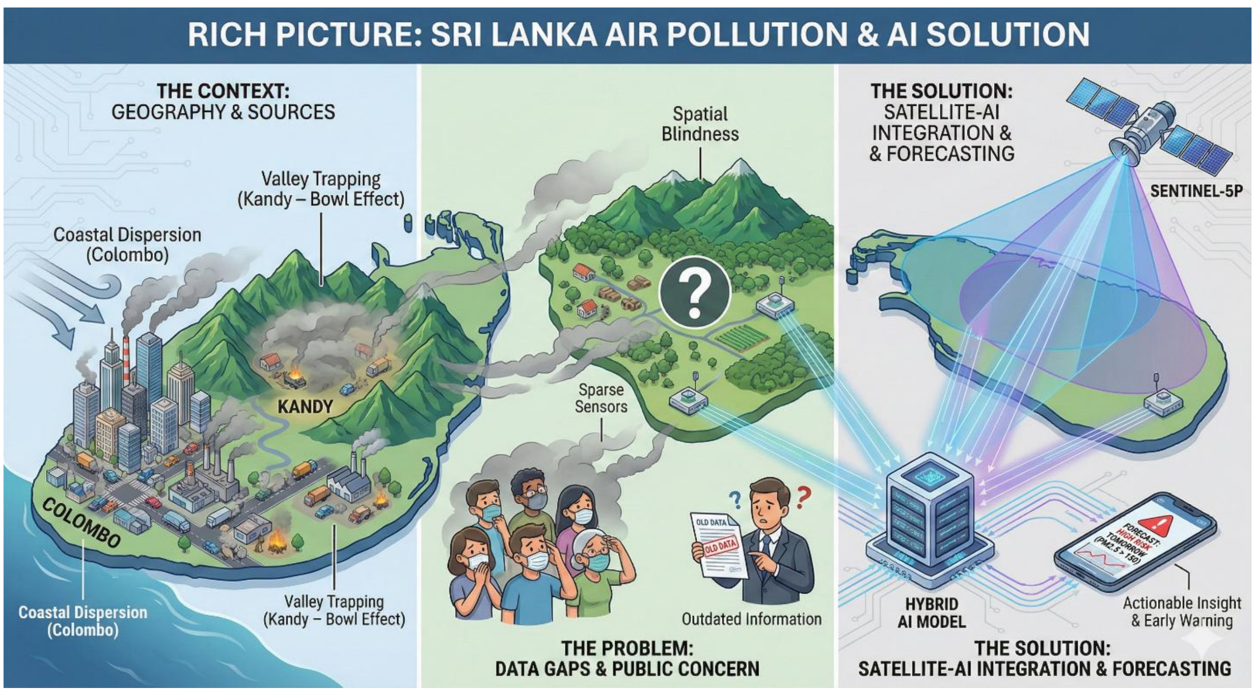

Figure 1 Rich picture

1.3 Problem Definition

1.3.1 Problem Statement

"Current air quality systems in Sri Lanka's primary metropolitan hubs, namely Colombo and Kandy, can only offer retrospective mapping and sparse monitoring systems, lacking a predictive system with the ability to forecast peaks in pollution (t+24h) and taking into account differences in local topographical characteristics (coastal vs. valley) and meteorological profiles (Ambanwala, 2025; Kurukulasuriya et al., 2025)

1.4 Key Related Work (Top 5)

The following table summarizes the five most important works that define the current gap this project is addressing.

Table 1.1: Comparative Analysis of Key Related Work

Citation (Author, Year, DOI or link)	**Contribution**	**Limitation**	**Gap Identified**	**Relevance to My Research**
Ambanwala (2025)	Analyzed pollution trends (1980–2024) in Colombo and Kandy using Modern-Era Retrospective analysis for Research and Applications (MERRA-2) data.	Retrospective Only. Provided historical analysis but no predictive model for future forecasting (t+1).	Forecasting Gap: No capability to predict future air quality.	Justifies my focus on Predictive Modeling for these t.
Kurukulasuriya et al. (2025)	Benchmarked Seasonal Auto-Regressive Integrated Moving Average (SARIMA) vs. Extreme Gradient Boosting (XGBoost) for Colombo AQI prediction.	Relied solely on ground history; ignored spatial context and regional transport.	Data Gap: Explicitly called for "hybrid modeling using satellite data."	I will implement the Hybrid Fusion requested by this paper.

Citation (Author, Year, DOI or link)	**Contribution**	**Limitation**	**Gap Identified**	**Relevance to My Research**
Attanayake et al. (2025) 10.1016/j.hazadv.2025.100782	Introduced satellite monitoring using PlanetScope imagery and Random Forest - Convolutional Neural Network (RF-CNN) pipelines.	Relied on Optical/Visual data (Land Use) which cannot detect invisible gas plumes from traffic.	Modality Gap: Visual imagery misses dynamic chemical changes (NO_2).	Justifies using Chemical Sensing (Sentinel-5P NO_2) as a feature.
Rowley & Karakuş (2023) 10.1016/j.rse.2023.113609	Proved that Multi-Modal AI (Satellite + Ground) outperforms uni-modal models globally.	Study focused on the UK; not calibrated for Sri Lanka's Monsoon & Valley dynamics.	Context Gap: Method needs adaptation for Tropical/Monsoon climates.	Provides the Methodological Foundation for my solution.
Dhammapala et al. (2022) https://doi.org/10.4209/aaqr.210266	Created the first static $PM_{2.5}$ map of Sri Lanka using Moderate Resolution Imaging Spectroradiometer (MODIS) AOD.	Focused on Static Mapping (Annual Averages) rather than daily dynamics.	Temporal Gap: Mapping tells us "where," not "when."	Justifies the need for High-Frequency Forecasting.

Table 1 Comparative analysis of Key Related Work

1.5 Research Gap

Gap 1: The "Spatio-Temporal" Gap (Uni-Modal vs. Multi-Modal Forecasting)

While prediction models for Sri Lanka exist, they are predominantly Uni-Modal and Spatially Blind. Studies such as Kurukulasuriya et al. (2025) and Fernando et al. (2022) successfully implemented XGBoost and Random Forest models for Colombo. However, these models rely exclusively on historical ground sensor data (Temporal). They lack Spatial Awareness—meaning they cannot detect incoming transboundary pollution (approaching Colombo from the ocean) or account for the specific "Valley Trapping" effect in Kandy (Ambanwala, 2025) until the pollution has already hit the sensor. There is no existing framework in Sri Lanka that fuses Spatial Satellite Data with Temporal Ground Data to predict *incoming* pollution events before they arrive.

Gap 2: The "Modality" Gap (Optical vs. Chemical Sensing)

Existing satellite studies for Sri Lanka (Attanayake et al., 2025) rely on optical land-use imagery (e.g., counting roads). This approach fails to capture dynamic emissions from traffic jams or industrial accidents that do not change the visual landscape. No study has fused Tropospheric Nitrogen Dioxide (NO_2) from Sentinel-5P, which serves as a direct proxy for traffic intensity, to model the distinct emission profiles of Colombo (high traffic volume) vs. Kandy (high congestion density).

Gap 3: The "Interpretability" Gap (Black Box AI)

Policymakers in these cities face different challenges: Colombo deals with transboundary smog, while Kandy deals with local trapping. Existing ML models (Random Forest, Support Vector Machines (SVM)) operate as "Black Boxes" and cannot distinguish between these causes. There is a complete absence of Explainable AI (SHAP) in the local literature to quantify whether a spike in Kandy is driven by local traffic or regional weather.

1.6 Contribution

1. Problem Domain (Practical Impact)

This project creates a "Virtual Sensor" capability by correlating satellite data with ground truth, the system can infer pollution levels in the suburbs of Colombo and the hills surrounding Kandy where no physical sensors exist. This allows for:

Targeted Interventions: Differentiating between days requiring traffic restrictions (High Local Emissions) vs. days requiring health advisories (High Regional Smog).

Cost Reduction: Expanding monitoring coverage without purchasing expensive hardware.

2. Research Domain (Algorithmic Novelty)

This project contributes a novel Hybrid Spatio-Temporal Architecture. Standard models process time-series data or spatial data in isolation. This project introduces a dual-branch neural network that fuses:

Temporal Data: Ground sensor history processed by LSTMs (Long Short-Term Memory).

Spatial Data: Satellite chemical vectors (NO_2, AOD) processed by Dense Layers.

This contributes a methodology for handling Multi-Modal Frequency Mismatch (aligning 10-minute ground data with daily satellite data) specifically calibrated for tropical urban environments.

3. To Body of Knowledge (Academic Advancement)

This research contributes empirical evidence to the global body of knowledge by validating the efficacy of satellite-based "Virtual Sensing" in tropical, topographically diverse regions.

Topographical Validation: It provides comparative analysis of how well satellite proxies perform in Coastal Flatlands (Colombo) versus Mountain Valleys (Kandy), addressing a gap in literature which mostly focuses on homogenous terrains.

Proxy Validation: It empirically quantifies the correlation between Tropospheric NO_2 and Ground-level PM_2.5 under Sri Lankan monsoon conditions.

1.7 Research Challenges

1. The Spatio-Temporal Misalignment Problem

Challenge: Ground sensors (PurpleAir) in Colombo/Kandy provide data every 10 minutes. The Sentinel-5P satellite passes overhead only once per day (approx. 13:30 LT).

Justification: Direct fusion is impossible without transformation. As noted by Kurukulasuriya et al. (2025), simple concatenation leads to dimensionality errors. The challenge lies in developing an interpolation strategy to "stretch" the daily satellite signal across 24 hours.

2. Topographical Complexity & Cloud Occlusion

Challenge: Kandy’s valley topography complicates satellite retrieval, and Sri Lanka’s tropical climate results in frequent cloud cover blocking MODIS/TROPOMI sensors.

Justification: Dhammapala et al. (2022) reported low data availability during monsoons. The challenge is to implement robust imputation techniques (e.g., KNN) to estimate missing values, ensuring the model remains accurate even when Kandy is under cloud cover.

3. The "Black Box" Trade-off

Challenge: Deep Learning models (LSTMs) achieve high accuracy but lack transparency.

Justification: For policy decisions in diverse cities (Closing schools in Kandy vs. restricting trucks in Colombo), authorities need to know the cause. Integrating SHAP (SHapley Additive exPlanations) is computationally expensive but necessary to deconstruct the model's decision-making.

1.8 Research Questions

RQ1: To what extent can the computational fusion of daily satellite chemical vectors (NO_2) with hourly ground data improve the RMSE of PM_2.5 forecasts in distinct urban topographies (Colombo vs. Kandy)?

RQ2: Does a Hybrid LSTM-Dense architecture demonstrate statistically significant superiority over traditional machine learning models (XGBoost) in capturing non-linear pollution trends in valley (Kandy) and coastal (Colombo) environments?

RQ3: What is the quantifiable variance in PM_2.5 levels attributed specifically to vehicular emissions NO_2 versus meteorological dispersion in these two cities, as isolated by SHAP analysis?

1.9 Research Aim

To design, develop, and evaluate a **Spatio-Temporal Hybrid Deep Learning Framework** that fuses ground-sensing, satellite imaging, and traffic proxies to forecast and explain $PM_{2.5}$ variations specifically for the urban centers of Colombo and Kandy.

1.10 Research Objectives

1. **Construct** a multi-modal dataset specifically for Colombo and Kandy combining PurpleAir history, Sentinel-5P (NO_2), and MODIS (AOD) data, applying cleaning techniques for cloud occlusion **by the end of the Data Collection phase (Week 14)**.

2. Design a Hybrid LSTM-Dense neural network to forecast PM_2.5 concentrations for a 24-hour horizon **during the Implementation phase (Weeks 15–20)**.

3. Evaluate the model against baseline approaches (ARIMA, XGBoost) using RMSE metrics, explicitly comparing performance in Coastal vs. Valley topographies **by the completion of Testing (Week 22)**.

4. Implement SHAP (SHapley Additive exPlanations) to visualize feature importance and provide decision support for urban planners **prior to drafting the final discussion (Week 24)**.

1.11 Chapter Summary

This chapter defined the critical "Spatio-Temporal Gap" in air quality monitoring for Sri Lanka's two major cities. It highlighted the distinct challenges of **Colombo (Coastal)** and **Kandy (Valley)**, identifying the limitations of current retrospective approaches. It proposed a Multi-Modal Data Fusion solution powered by Deep Learning to bridge this gap. The following chapter provides a comprehensive review of the literature supporting this approach.

CHAPTER 2: LITERATURE REVIEW

2.1 Chapter Overview

This chapter provides a comprehensive and critical analysis of the theoretical and empirical foundations governing air quality forecasting in tropical urban environments. It moves beyond the general problem statement to deconstruct the specific atmospheric dynamics of Sri Lanka's two distinct economic hubs: the coastal flatlands of **Colombo** and the mountain basin of **Kandy**. The review is structured thematically to bridge the gap between Environmental Science and Computer Science.

Section 2.3.1 examines the **Problem Domain**, contrasting the dispersion physics of coastal winds against valley trapping mechanisms. Section 2.3.2 provides a **Technological Review**, evaluating the shift from ground-based IoT sensing to satellite-based "Virtual Sensing," with a specific focus on the physics of the **Sentinel-5P TROPOMI** instrument. Section 2.3.3 critiques the evolution of forecasting algorithms, detailing the mathematical limitations of statistical models (ARIMA) versus the capabilities of Deep Learning (LSTM). Finally, Section 2.4 establishes the quantitative **Evaluation Framework**, defining the specific metrics required to benchmark the proposed hybrid architecture against existing solutions.

2.2 Concept Map

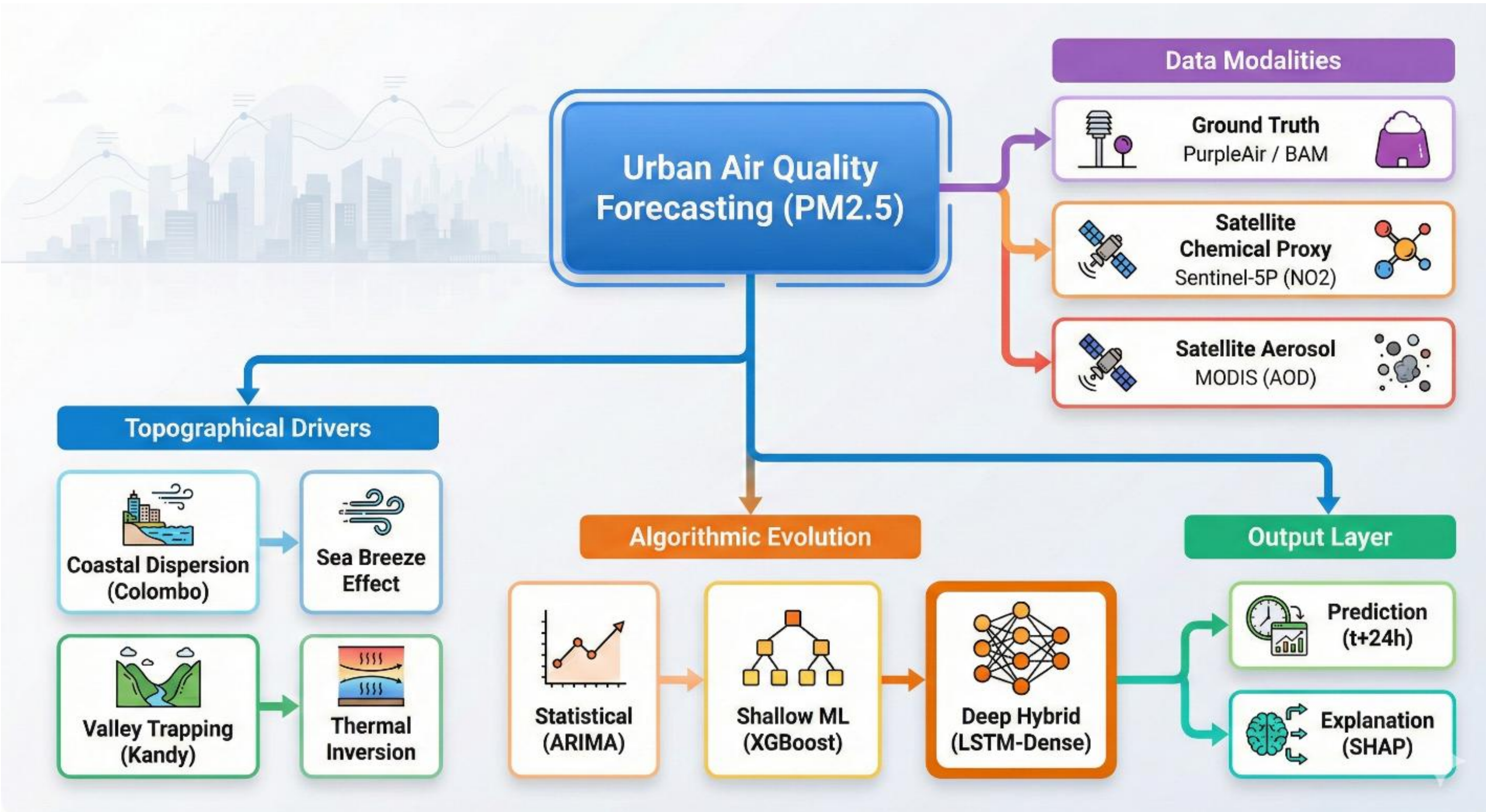

Figure 2 Concept map

2.3 Thematic Review

2.3.1 Problem Domain: The "Tale of Two Cities"

Air quality in Sri Lanka is not a homogenous phenomenon; it is a complex derivative of **Anthropogenic Emissions**, **Monsoonal Meteorology**, and most critically, **Local Topography**.

1. The Coastal Dynamics of Colombo

Colombo, situated in the Western Province, represents a classic coastal urban environment.

- **Emission Sources:** As the commercial capital, it hosts the highest density of vehicular traffic in the country. **Rathnayake et al. (2023)** note that 60% of Sri Lanka's vehicle fleet operates within the Western Province, leading to high chronic emissions of Nitrogen Oxides (NO_x) and Particulate Matter.
- **Dispersion Mechanisms:** Despite high emissions, Colombo benefits from the **"Sea Breeze Effect."** Daytime heating of the land creates a pressure differential that draws clean marine air into the city, naturally diluting pollutants. However, this mechanism fails during the **Northeast Monsoon (Dec–Feb)**. **Ambanwala (2025)** provides longitudinal evidence showing that during this season, wind patterns reverse, carrying transboundary pollution plumes from the Indian subcontinent across the ocean, causing city-wide smog events that local sensors often misinterpret as local traffic spikes.

2. The Valley Dynamics of Kandy

In stark contrast, Kandy represents a "Bowl Effect" topography. Situated 500 meters above sea level and surrounded by the Hanthana and Knuckles mountain ranges, Kandy suffers from critically poor ventilation.

- **Pollutant Trapping: Ambanwala (2025)** confirmed that while Kandy has fewer vehicles than Colombo, it frequently records higher $PM_{2.5}$ concentrations. This is attributed to the city's geomorphology, which prevents the horizontal dispersion of pollutants.
- **Thermal Inversion:** During colder months, Kandy experiences **Thermal Inversion**, where a layer of warm air sits atop a layer of cold air near the ground. This "lid" traps vehicle exhaust in the valley floor, causing pollutants to accumulate cumulatively over

days. Existing statistical models fail to capture this accumulation effect, often under-predicting toxicity levels in Kandy during the early morning hours.

2.3.2 Technological Review

1. Data Acquisition: The Physics of Virtual Sensing

- **Ground Sensing:** Traditional monitoring relies on **Beta Attenuation Monitors (BAM)**, which measure the absorption of beta radiation by particles collected on a filter tape. While highly accurate, they are expensive ($20k+) and sparse. **Dissanayaka et al. (2019)** attempted to bridge this gap with IoT solutions (Air Visio), but low-cost sensors suffer from calibration drift in high humidity.
- **Satellite Aerosol Sensing (MODIS):** The Moderate Resolution Imaging Spectroradiometer (MODIS) measures **Aerosol Optical Depth (AOD)**. AOD is a dimensionless measure of how much direct sunlight is prevented from reaching the ground by aerosols. While **Dhammapala et al. (2022)** successfully used AOD for mapping, it has a limitation: it measures *total* light extinction, making it difficult to distinguish between dust, smoke, or industrial pollution.
- **Satellite Chemical Sensing (Sentinel-5P):** The critical advancement for this project is the **Tropospheric Monitoring Instrument (TROPOMI)** on the Sentinel-5P satellite. Unlike MODIS, TROPOMI is a spectrometer that detects the specific absorption fingerprints of trace gases. It specifically measures **Tropospheric Nitrogen Dioxide (NO_2)**. **Holloway et al. (2021)** established that NO_2 is a short-lived pollutant (lifetime < 24 hours) primarily emitted by high-temperature combustion (vehicles). Therefore, satellite-detected NO_2 serves as a near-perfect dynamic proxy for **urban traffic intensity**, allowing the model to "see" congestion in Kandy or Colombo even without ground cameras.

2. Algorithmic Review: From Linearity to Deep Memory

- **Statistical Models (ARIMA):** Early studies, such as **Kurukulasuriya et al. (2025)**, utilized **ARIMA (Auto-Regressive Integrated Moving Average)**. These models rely on the assumption that future values are a linear function of past values. However, the study concluded that ARIMA models struggle with the chaotic, non-linear nature of air pollution, resulting in a high RMSE (16.99).
- **Shallow Machine Learning:** Studies by **Gupta et al. (2023)** and **Mihirani et al. (2023)** improved upon this using **Random Forest** and **Support Vector Machines (SVM)**. These algorithms can model non-linear relationships but treat time-series data as independent tabular rows. They lack an internal state to "remember" that it rained three days ago, which is crucial for predicting soil dust resuspension.
- **Deep Learning (LSTMs):** To address the limitations of "memoryless" models, global literature has shifted to **Long Short-Term Memory (LSTM)** networks. LSTMs solve the "Vanishing Gradient Problem" of standard RNNs using a gating mechanism:
    - **Forget Gate:** Decides what information (e.g., old seasonal trends) to discard.
    - **Input Gate:** Decides what new information (e.g., today's traffic spike) to store.
    - **Cell State:** Acts as a conveyor belt, carrying relevant information across long time sequences.

This architecture is theoretically ideal for Kandy, where the *accumulation* of pollution over several days is a key predictor

2.3.3 Comparative Analysis of Approaches

Approach	Representative Work	Strengths	Weaknesses	Relevance to Project
Statistical	Kurukulasuriya et al. (2025) (SARIMA)	Computationally cheap; good for seasonal averages.	Fails to predict sudden spikes; low accuracy (RMSE ~16.99).	Serves as the **Baseline** to beat.
Optical Satellite ML	Attanayake et al. (2025) (RF-CNN)	Covers wide areas; uses visual land features.	Cannot detect dynamic gas emissions; relies on static maps.	Highlights the need for **Chemical (NO_2)** data.
Hybrid Deep Learning	Rowley et al. (2023) (Global)	High accuracy; fuses Satellite + Ground data.	Complex to architect; rarely applied to **Tropical/Monsoon** climates.	Serves as the **Proposed Architecture**.

Table 2 Comparative analysis of approaches

2.4 Evaluation and Benchmarking

To ensure the proposed solution is academically robust, it must be evaluated against established quantitative metrics.

1. Quantitative Metrics

- **RMSE (Root Mean Square Error):**

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Figure 3 RMSE formula

RMSE is selected as the primary metric over Mean Absolute Error (MAE) because it penalizes large errors more heavily. In the context of air quality, missing a hazardous smog event (a large error) has severe health consequences, making RMSE the safety-critical metric of choice.

- **R^2 (Coefficient of Determination):** Measures the proportion of the variance in the dependent variable (PM_2.5) that is predictable from the independent variables (Satellite/Weather).

2. Benchmarks for Success

Based on the literature review, the project aims to exceed the following benchmarks:

- **Baseline RMSE:** The project aims to achieve an RMSE lower than **16.99**, which was the best result achieved by the SARIMA model in **Kurukulasuriya et al. (2025)**.
- **Baseline R^2:** The project targets an R^2 > 0.85, improving upon the **0.72** achieved by Random Forest in **Kekulanadara (2021)**.

2.5 Chapter Summary

This chapter critically reviewed the atmospheric dynamics of Colombo and Kandy, validating the need for a spatially-aware forecasting model. It established that while satellite data has been used for *retrospective mapping* (Dhammapala) and *visual estimation* (Attanayake), there is a significant gap in **predictive chemical forecasting**. The review of algorithmic approaches confirmed that **Hybrid LSTM** architectures offer the theoretical capacity to solve the "Memory" and "Non-linearity" problems identified in current Sri Lankan studies. This theoretical foundation informs the Methodology detailed in Chapter 3.

CHAPTER 3: METHODOLOGY

3.1 Chapter Overview

This chapter delineates the rigorous methodological framework adopted to engineer the proposed **Spatio-Temporal Air Quality Forecasting System**. It is structured into two distinct methodological streams: the **Research Methodology**, which governs the scientific inquiry based on **Saunders' Research Onion**, and the **Development Methodology**, which governs the software engineering lifecycle using **Agile Prototyping**. The chapter details the specific data fusion strategies for Colombo and Kandy, the neural network architecture, and the quantitative evaluation protocols required to validate the research hypotheses.

3.2 Research Methodology (The Onion)

To ensure academic rigor, this research adopts the layered approach defined by the **Research Onion** (Saunders et al., 2019).

1. Philosophy: Positivism

- **Choice:** Positivism.
- **Justification:** This research operates on the premise that air quality is an objective, measurable physical reality governed by discoverable laws (e.g., fluid dynamics, chemical transport). The variables involved—$PM_{2.5}$ concentration ($\mu g/m^3$), Aerosol Optical Depth (AOD), and Wind Speed (m/s)—are quantifiable and independent of the observer. The study aims to empirically test the correlation between satellite proxies and ground truth, devoid of subjective interpretation.

2. Approach: Deductive

- **Choice:** Deductive.
- **Justification:** The project does not begin with observation to form a theory (Inductive). Instead, it begins with a theoretical hypothesis derived from global literature (Rowley & Karakuş, 2023): *"Fusing satellite chemical data with ground sensors reduces forecasting error in urban environments."* The research is designed to test this hypothesis specifically within the context of Sri Lankan topography.

3. Methodological Choice: Mono-Method Quantitative

- **Choice:** Mono-Method Quantitative.
- **Justification:** The study relies exclusively on numerical data analysis. It involves the statistical processing of large-scale environmental datasets (Time-Series and Spatial Vectors) and the evaluation of model performance using quantitative metrics (RMSE, R^2). No qualitative data (e.g., interviews/surveys) is required to answer the research questions.

4. Strategy: Experiment

- **Choice:** Controlled Experiment.
- **Justification:** The core strategy is to conduct a comparative performance analysis. The **Independent Variable** is the input feature set (Uni-modal Ground Data vs. Multi-modal Satellite Fusion), and the **Dependent Variable** is the forecasting accuracy. The experiment compares the Proposed Model against a Baseline Model (XGBoost) under controlled conditions (same timeframe, same test set).

5. Time Horizon: Longitudinal

- **Choice:** Longitudinal.
- **Justification:** Air quality is inherently seasonal. A "Snapshot" (Cross-sectional) analysis would fail to capture the critical influence of the Northeast Monsoon (Dec–Feb) vs. the Southwest Monsoon (May–Sep). Therefore, this study analyzes historical data spanning **2018–2024** to capture long-term trends and cyclical variations.

6. Data Collection & Analysis

- **Population:** The atmospheric column over the Western Province (Colombo) and Central Province (Kandy).
- **Sample:** A continuous 6-year time series of hourly ground readings and daily satellite swathes.
- **Tools:** Python (Pandas, TensorFlow/Keras), Google Earth Engine (GEE), SHAP Library.

3.3 Development Methodology (Agile)

Given the exploratory nature of Deep Learning—where hyperparameters and architecture depths are not known a priori—a linear "Waterfall" model is unsuitable.

- **Method: Agile Prototyping**.
- **Justification:** The project requires iterative cycles of "Train-Evaluate-Tune." An initial prototype (Baseline Model) will be built in the first sprint to establish a performance floor. Subsequent sprints will incrementally add complexity (adding the LSTM branch, then the Satellite branch, then the Explanation layer). This allows for rapid failure identification (e.g., if a specific satellite product is too noisy) and adaptatio

3.3.1 Solution Methodology

Step 1: Dataset Selection The following datasets have been selected to construct the **"Virtual Sensor"** network.

Dataset	Size & Type	Features/Attributes	Modality	Licensing & Access	Limitation	Suitability for Project
PurpleAir (Ground Truth)	~200k rows (Hourly)	PM_2.5 (CF=1), Temp, Humidity	Time-Series (Tabular)	Public Domain (API)	Sparse coverage; sensor drift	**High.** Provides the "Label" (y) for supervised learning. Validated by Dhammapala et al. (2022).
Sentinel-5P TROPOMI	~500 GB (Daily Swathes)	Tropospheric NO_2 Column Density	Spatial Raster (Image)	Open Access (ESA Copernicus)	Cloud cover occlusion; daily frequency	**High.** Validated proxy for urban traffic intensity in Colombo/Kandy.

Dataset	Size & Type	Features/Attributes	Modality	Licensing & Access	Limitation	Suitability for Project
MODIS (Terra/Aqua)	~300 GB (Daily Swathes)	Aerosol Optical Depth (AOD) 550nm	Spatial Raster (Image)	Open Access (NASA)	Low resolution (10km); cannot distinguish pollutant types	**Medium.** Useful for detecting regional background smog (transboundary).
OpenMeteo Historical	~200k rows (Hourly)	Wind Speed (10m/100m), Direction, Rain	Time-Series (Tabular)	Open Access (CC-BY 4.0)	Reanalysis data (not direct observation)	**High.** Essential for modeling dispersion (Wind) and washout (Rain).

Table 3 Dataset selection

Step 2: Preprocessing & Feature Engineering

1. **Temporal Alignment:** Ground data is hourly; Satellite data is daily (approx. 13:30 LT).
    - *Technique:* **Linear Interpolation**. We assume the "Regional Background" (NO_2/AOD) changes slowly over 24 hours, creating a continuous context vector.
2. **Imputation (Handling Cloud Cover):** Kandy is frequently cloudy.
    - *Technique:* **KNN Imputation**. Missing AOD pixels will be estimated based on the values of the nearest clear pixels and the previous day's value.
3. **Feature Engineering:** Creation of Lag Features (t_-1, t_-2... t_-24) to give the LSTM "memory" of the previous day's pollution profile.

Step 3: Model Selection & Architecture

- **Architecture:** Hybrid Multi-Input Neural Network.
    - *Input A:* Sequence Data -> **LSTM Layers** (captures temporal trends).
    - *Input B:* Satellite Vector -> **Dense Layers** (captures spatial context).
    - *Fusion:* Concatenation Layer -> **Regression Head**.

Step 4: Training & Validation

- **Split Strategy: Time-Series Split** (Walk-Forward Validation). Random shuffling is forbidden as it would leak future information into the past.
    - *Train:* Jan 2018 – Dec 2022.
    - *Test:* Jan 2023 – Dec 2024 (unseen data).

3.3.2 Evaluation Methodology

The solution will be evaluated against the "Black Box" and "Uni-modal" baselines identified in the Literature Review.

1. Quantitative Metrics

- **RMSE (Root Mean Square Error):** The primary metric. It penalizes large errors heavily, which is critical for safety (e.g., missing a toxic smog event is worse than a small error on a clean day).
    - *Target:* RMSE < 15.0 (Beating the 16.99 baseline set by Kurukulasuriya et al., 2025).
- **R^2 (Coefficient of Determination):** To measure how well the model replicates the variance of the real data.
    - *Target:* R^2 > 0.85.

2. Qualitative Analysis (Explainability)

- **SHAP Summary Plots:** Will be generated to visualize feature importance.
    - *Success Criterion:* The model must show that **NO_2 (Traffic)** is a top-3 contributor to PM_2.5 spikes in Colombo during weekdays, validating the physical consistency of the model.

3.4 Project Management

3.4.1 Scope

- **In-Scope:**
  - Forecasting PM_2.5 for Colombo and Kandy.
  - Data period: 2018–2024.
  - Development of the Hybrid Deep Learning Model.
  - SHAP-based analysis.

- **Out-of-Scope:**
  - Forecasting other gases (O_3, SO_2) unless used as features.
  - Covering whole of Sri Lanka (Only Colombo and Kandy).

3.4.2 Gantt Chart

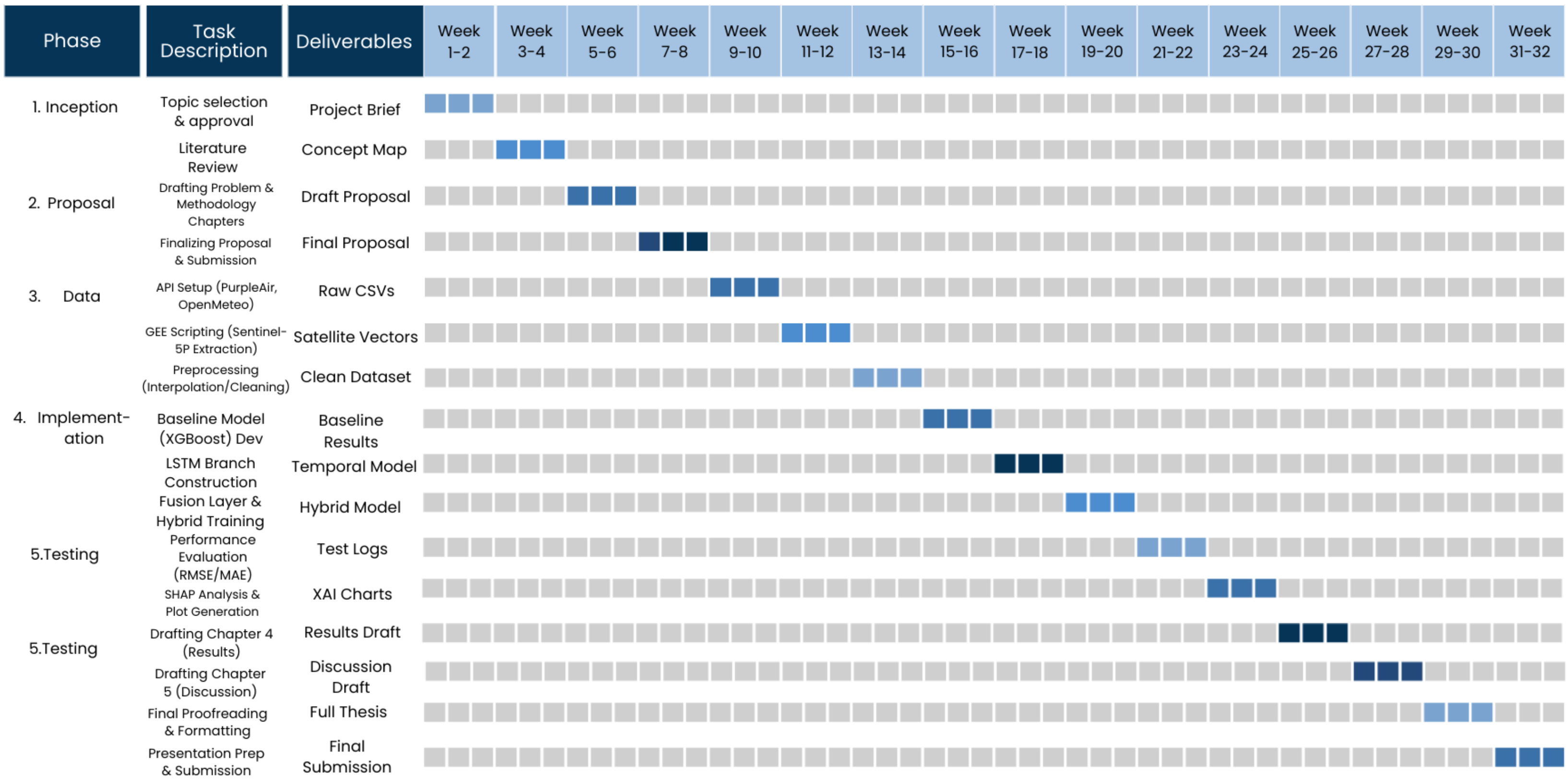

Figure 4 Gantt chart

3.4.3 Resource Requirements

- **Hardware:** Google Colab Pro (NVIDIA T4/A100 GPU) for accelerated model training.
- **Software:** Python 3.9, TensorFlow, Google Earth Engine (GEE) API.
- **Data:** Satellite archive (processed cloud-side via GEE).

3.4.4 Risk Analysis & Mitigation

Risk	Impact	Likelihood	Mitigation Strategy
Cloud Cover in Kandy (Satellite Blindness)	High	High	Primary: Use KNN Imputation to estimate missing pixels based on neighbors. Fallback: Implement dynamic weight adjustment to prioritize the LSTM (Ground History) branch when satellite data confidence drops below a set threshold.
Google Earth Engine (GEE) Computation Limits	Medium	Medium	Export satellite data in annual batches to Google Drive rather than processing on-the-fly.
Model Overfitting	High	Medium	Implement Early Stopping and Dropout Layers (0.2) in the LSTM architecture.
PurpleAir Sensor Drift	Medium	Medium	Apply the calibration factors derived by Dhammapala et al. (2022) during preprocessing.

Table 4 Risk table

3.5 Ethics & Compliance Checklist

Ethical Concern	Action Taken	Evidence / Documentation
Informed Consent	N/A (No human participants).	N/A
Data Privacy	No PII processed. While PurpleAir sensor locations are public, coordinates of residential sensors will be rounded to 3 decimal places (approx. 110m) to prevent exact home identification in the final dataset.	Data Management Plan included in proposal.
Bias & Fairness	Spatial Bias: Acknowledged that sensors are in wealthy areas.	The "Limitations" section explicitly states the model may be less accurate for low-income suburbs.
Licensing	All satellite data is Open Access (Copernicus/NASA).	GEE Terms of Service attached in Appendix.

Table 5 Ethics table

3.6 Chapter Summary

This chapter detailed the research design, justifying the Positivist/Quantitative approach to modeling atmospheric physics. It outlined the Agile development of a Hybrid LSTM-Dense architecture and established a rigorous validation framework using RMSE and SHAP. The specific challenges of cloud cover in Kandy were addressed via imputation strategies, ensuring the methodology is robust for the Sri Lankan context.

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